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TRAVELING SALESMAN-TYPE COMBINATORIAL
PROBLEMS AND THEIR RELATION TO THE
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PREVIEW

NORTHWESTERN UNIVERSITY

TRAVELING SALESMAN-TYPE COMBINATORIAL PROBLEMS AND
THEIR RELATION TO THE LOGISTICS OF REGIONAL BLOOD BANKING

A DISSERTATION

SUBMITTED TO THE GRADUATE SCHOOL
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS

for the degree

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and Management Sciences

By

ILHAN OR

Evanston, Illinois

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ABSTRACT

Four traveling salesman type problems have been investigated, and formulated as integer programming problems. A new nonlinear programming formulation has been presented for the Traveling Salesman Problem, and through this formulation an interesting relationship between the traveling salesman problem and posynomial programming has been developed. A very fast and quite accurate new heuristic algorithm has been presented for the traveling salesman problem. Another similar algorithm has been developed for the Multi-depot Traveling Salesman Problem. A relationship between traveling salesman type very general and complex transportation problems and the simpler location allocation problem and the transportation problem has been presented.

The logistics of regional blood banking have been studied and a realistic model has been developed. An integer programming formulation for this model has been presented. This integer programming problem, which is a complex combination of the traveling salesman problem and the location-allocation problem, has been investigated and solution procedures for it, based on the algorithms we have developed for the traveling salesman problem and the multi-depot traveling salesman problem, have been presented.

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1. Introduction

The research for this dissertation has been generated while investigating the transportation aspects of the regional blood banking systems. During this investigation two well-known, complex, combinatorial problems, namely the Vehicle Dispatch Problem and the Traveling Salesman Problem, and their generalizations, namely the Multi-Depot Vehicle Dispatch Problem and the Multi-Depot Traveling Salesman Problem, have been studied in detail. New heuristic solution procedures have been developed for the traveling salesman problem and the multi-depot traveling salesman problem. A general transportation problem which includes the traveling salesman problem and its generalization, the vehicle dispatch problem and its generalization, the transportation problem, the location-allocation problem and the transportation-location problem as special cases has been formulated and studied. A model, focusing on the transportation aspects of the regional blood banking systems has been constructed; then, using the results of the study on the general transportation problem, the related optimization problem, to the model we have constructed, has been formulated (this problem is called the blood transportation-allocation problem). Finally, using the solution procedures developed for the traveling salesman problem and the multi-depot traveling salesman problem, along with an existing algorithm for the vehicle dispatch problem, solution procedures for the blood transportation-allocation

problem have been developed.

In chapter 2 the general transportation problem (GTP) is formulated. This problem may informally be stated as "N demand points are to be supplied a single commodity by M supply points, through at most n supply vehicles, regularly. Given certain vehicle capacities and limitations, supply point capacities, demand point requirements and the set of available sites for the supply points, find the locations for the supply points, and determine the supply routes that will minimize the total distance travelled by the supply vehicles." Then, some simple but important results on the feasible and the optimal solution sets of the GTP are presented. Next, some special cases of the GTP are examined. First, the vehicle dispatch problem (VDP) is formulated and discussed. It is shown that the GTP reduces to the VDP when there is a single supply point and its location is given. Then, some existing solution procedures for the VDP are presented. The traveling salesman problem (TSP) is another important special case of the GTP. However, since the whole third chapter is devoted to this problem, in chapter 2 it is just shown that the GTP reduces to the TSP when there is a single supply point with a given location and there are no binding constraints on the vehicles. Next, the multi-depot problems, namely the multi-depot traveling salesman problem (MTSP) and the multi-depot vehicle dispatch problem (MVDP) are formulated as special cases of the GTP. Some of the existing solution procedures to these problems are discussed. However, the new heuristic solution procedure we have developed for the MTSP is not presented in this chapter but left to chapter 3, since it is based on our algorithm for the TSP, which is also

presented in chapter 3. Finally, the location-allocation problem (LAP), the transportation problem (TP), and the transportation-location problem (TLP) are presented as special cases of the GTP. It is shown how the GTP reduces to the TP when all vehicle capacities are equal and the requirements of demand points and the capacities of the supply points are all integer multiples of the vehicle capacities.

The transportation problems we are most interested in, namely the GTP, the VDP, the TSP, the MVDP and the MTSP, have a very important common feature. The deliveries of the goods are made in a traveling salesman fashion; in other words, the delivery vehicles do not return to the base after each delivery, but continue making other deliveries until some capacity or time limit is reached. Hence, the traveling salesman problem emerges as an important underlying subproblem in every problem discussed in this research. Chapter 3 is devoted to this major subproblem. In this chapter, the TSP is formulated, discussed and investigated in detail. In one section in this chapter, an interesting relationship between the TSP and posynomial programming is developed. It is shown that each traveling salesman problem is the limit of a certain sequence of posynomial programming problems and that there exists posynomial programming problems whose optimal solution sets will uniquely determine the optimal solution set of the traveling salesman problem under consideration. Unfortunately, all of the posynomial programming problems related to the TSP belong to a less developed area of posynomial programming. They are posynomial programming problems with reverse constraints; good solution procedures have not yet been found for these types of problems of any significant size. Hence, the

relationship we have shown would not yet help us in solving a TSP.

In another section in this chapter a fast new heuristic algorithm for the traveling salesman problem is presented. This algorithm, called the convex hull algorithm, is a tour building algorithm which makes extensive use of the topological considerations in a given TSP data. The algorithm starts with an initial subtour consisting of the boundary of the convex hull of the points in the given data. Then interior points are included to this subtour, one by one, in a certain order. A new heuristic algorithm for the MTSP, based on the convex hull algorithm is also presented in this chapter.

Chapter 4 is devoted to investigating the transportation aspects of the regional blood banking systems. Regionalization is a process by which a set of blood banks within a given geographical area moves towards the coordination of its activities. In our regionalization model only a two level hierarchy is considered. The central blood banks are on one level and the local blood banks on the other. Each local bank is to be assigned to one central bank. A central bank with all of the local banks assigned to it constitutes a single region that is to be coordinating its activities. The transportation aspects that are investigated in this chapter are central bank locations, regional boundaries and optimal distribution and communication network configurations. Our objective is to decrease the operating costs without sacrificing blood quality, research and education. The problem which is briefly and vaguely described above is built into a complete model in chapter 4 and then formulated as a mathematical programming problem. Then, heuristic solution procedures (in which the algorithms discussed

in the previous chapters are used extensively) for this problem are presented. Some of the results obtained by coding and applying these procedures on a CDC 6400 computer, are also presented in this chapter.

Chapter 5 contains some conclusions and suggestions for future research in related areas.

It should be noted that, since this research involved many distinct problems, the literature survey is not lumped together in any specific section, instead it is scattered throughout the dissertation and presented when it is relevant. Also, all of the new heuristic algorithms presented in this research have been coded in fortran IV and tested on a CDC 6400 computer. The actual codings and their descriptions may be obtained by contacting either the author or Professor William Pierskalla, Northwestern University, Evanston, Illinois.

2. A General Transportation Problem (or a Family of Combinatorial Problems)

2.1 Introduction and Problem Description

In this chapter we will be examining the following very general and loosely stated problem, and its interesting (and sometimes well known) special cases: "N demand points are to be supplied a single commodity by M supply points through at most n supply vehicles, regularly. Find the locations for the supply points and determine the supply routes that will minimize the total distance travelled by the supply vehicles." There can be many different types of constraints for this problem through capacities of supply vehicles; capacities of supply points; requirements of demand points; maximum distance travelled by supply vehicles (or maximum number of stops per supply vehicle); number of supply vehicles and locations available for supply points.

Let us now define the notation that will be used in discussing the above problem.

- i) "N" is the number of demand points.
- ii) "M" is the number of supply points.
- iii) "n" is the maximum number of supply vehicles available.
- iv) $\mathcal{D} = \{H_1, \dots, H_N\}$ is a set of N demand points.
- v) $\mathcal{S} = \{H_{N+1}, \dots, H_{N+M}\}$ is a set of M supply points.
- vi) (X_i, Y_i) , $i = 1 \dots N$ are the cartesian coordinates of the i^{th} demand point.

- vii) (X_i, Y_i) , $i = N + 1, \dots, N + M$ are the cartesian coordinates of $(i - N) = j^{\text{th}}$ supply point.
- viii) $d[(X_{i_1}, Y_{i_1}), (X_{i_2}, Y_{i_2})]$ $i_1 = 1 \dots N + M$; $i_2 = 1 \dots N + M$ is the distance between points H_{i_1}, H_{i_2} . Let "d" be a metric.
- ix) C_k , $k = 1 \dots n$ is the capacity of supply vehicle k .
- x) Q_i , $i = 1 \dots N$ is the requirement of demand point i .
- xi) S_j , $j = 1 \dots M$ is the capacity of supply point j ; or equivalently Q_i , $i = N + 1, \dots, N + M$ is the capacity of point i .
- xii) D_k , $k = 1 \dots n$ is the maximum distance supply vehicle k may travel.
- xiii) ω is the set of all feasible supply point locations.
- xiv) $\mathcal{N} = \mathcal{D} \cup \mathcal{S}$ is the set of all points involved in the problem.
- xv) $q = \{q_{ik}\}$ $i = 1 \dots N$; $k = 1 \dots n$ is the amount supplied to point i by vehicle k .

DEFINITION 2.1: A graph $g \equiv g(V, E)$ consists of a set of objects $V = \{v_1, v_2, \dots\}$, called vertices (points), and another set $E = \{e_1, e_2, \dots\}$ whose element are called edges, such that each edge e_k is identified with a pair (v_i, v_j) of vertices. The vertices v_i, v_j associated with edge e_k are called the end vertices of e_k .

When a vertex v_i is an end vertex of some edge e_j , v_i and e_j are said to be incident with each other. Two edges are said to be adjacent if they are incident on a common vertex. Similarly, two vertices are said to be adjacent if they are the end vertices of the same edge. The number of edges incident on a vertex (point) v_i , is called the degree, $dg(v_i)$, of vertex v_i . A walk is defined as a finite alternating

sequence of vertices and edges, such that each edge is incident with the vertices preceeding and following it. If a walk begins and ends at the same vertex it is called a closed walk. A walk in which no vertex is traversed more than once is defined to be a path. A closed walk in which no vertex (except the initial and the final vertex) appears more than once is called a circuit. A connected graph is a graph in which there is at least one path between every pair of its vertices. A closed walk running through every edge of a graph is called an Euler line, and a graph that consists of an Euler line is called an Euler graph. Through some basic results in graph theory [DEO, 1974] it is also possible to define Euler graph and circuit as follows:

DEFINITION 2.2: An Euler graph is a connected graph in which all vertices are of even degree.

DEFINITION 2.3: An Euler graph in which all vertices have degree 2 is called a circuit.

xvi) Let $\mathcal{L} = \{L_1, \dots, L_n\}$ where $L_i \equiv L_i(V_i, E_i)$ is an Euler graph, $V_i \subset \mathcal{N}$, and it is the route travelled by vehicle i .

Now, with this notation and these definitions, we can formally state the General Transportation Problem (GTP):

$$(2.1) \min z[(X, Y), \mathcal{L}, q] = \sum_{\{j: L_j \in \mathcal{L}\}} \sum_{\{(i_1, i_2): (H_{i_1}, H_{i_2}) \in L_j\}} d[(X_{i_1}, Y_{i_1}), (X_{i_2}, Y_{i_2})]$$

subject to

- i) $q_{ik} = 0$ if $H_i \notin L_k$ or $L_k \cap \mathcal{D} = \emptyset$ $H_i \in \mathcal{D}$, $L_k \in \mathcal{L}$
- ii) $\sum_{k=1}^n q_{ik} \geq Q_i \quad i = 1, \dots, N$
- iii) $\sum_{i=1}^N q_{ik} \leq C_k \quad k = 1, \dots, n$
- iv) $\sum_{k \in K_j} \left(\sum_{i=1}^N q_{ik} \right) \leq S_j \quad j = 1, \dots, M; K_j = \{k: H_{N+j} \in L_k\}$
- v) $\sum_{\{(i_1, i_2): (H_{i_1}, H_{i_2}) \in L_k\}} d[(X_{i_1}, Y_{i_1}), (X_{i_2}, Y_{i_2})] \leq D_k \quad k = 1, \dots, n$
- vi) $(X_i, Y_i) \in \omega \quad H_i \in \mathcal{D}$

In the above problem we have three types of variables:

$\{(X_i, Y_i) \quad i = N + 1, \dots, N + M\}$ are unrestricted real variables in R^2
 $\{q_{ik} \quad i = 1, \dots, N; k = 1, \dots, n\}$ are nonnegative real variables in R^1

$\mathcal{L}$ is any collection of n Euler graphs. (Note that there can be only finitely many different $\mathcal{L}$).

Constraints (i) mean that an Euler graph k corresponds to a supply vehicle k only if it has a supply point in it ($L_k \cap \mathcal{D} \neq \emptyset$) and that a supply vehicle, k , cannot satisfy any of the demand of point i if it is not going there ($H_i \notin L_k$).

Constraints (ii) mean that supply vehicles corresponding to the first n Euler graphs must satisfy the demand of demand point H_i , $i = 1, \dots, N$.

(Requirement constraints)

Constraints (iii) mean that the total amount of goods carried by vehicle k must be less than or equal to the capacity of vehicle k , $k = 1, \dots, n$. (Vehicle capacity constraints)

Constraints (iv) mean that the total amount of goods supplied by supply point j must be less than or equal to the capacity of supply point j , $j = 1, \dots, M$ (Supply Point capacity constraints)

Constraints (v) mean that the total distance travelled by supply vehicle k must be less than or equal to the maximum allowed for vehicle k , $k = 1, \dots, n$. (Maximum Travel Distance constraints)

Constraints (vi) give the set of feasible locations for supply points.

Note that the above problem is not feasible unless

$$\sum_{j=1}^M s_j \geq \sum_{i=1}^N Q_i \quad \text{and} \quad (1)$$

$$\sum_{k=1}^n c_k \geq \sum_{i=1}^N Q_i \quad (2)$$

The GTP is a very complex and extremely difficult problem to solve. Its special cases cover a broad range of well known transportation related problems, some of which we will examine in detail later. Now, let us take a closer look at the optimal and feasible solution sets of the GTP.

2.2 Some Results About the Feasible Solution Set

LEMMA 2.1: Let $[(X^0, Y^0), \mathcal{L}^0, q^0]$ be any feasible solution to problem (2.1). $(X^0, Y^0) = \{(X_i^0, Y_i^0) \mid i = 1, \dots, N + M\}$. Then, there exists

$[(X^0, Y^0), \mathcal{L}^*, q^0]$ such that for all $L_k^* \in \mathcal{L}^*$ either $L_k^* = \emptyset$ or $|L_k^* \cap \mathcal{L}^0| = 1$ (3) and $z[(X^0, Y^0), \mathcal{L}^*, q^0] \leq z[(X^0, Y^0), \mathcal{L}^0, q^0]$ with $[(X^0, Y^0), \mathcal{L}^*, q^0]$ still feasible.

PROOF: Assume for $L_{k_1}^0 \in \mathcal{L}^0$ $|L_{k_1}^0 \cap \mathcal{S}| = 2$ and for $L_{k_2}^0 \in \mathcal{L}^0$, $L_{k_2}^0 \cap \mathcal{S} = \emptyset$ with $L_{k_2}^0 \neq \emptyset$. Then there exists $H_{i_1}, H_{i_2} \in L_{k_1}^0 \cap \mathcal{S}$, $i_1 \neq i_2$ and $H_{j_1}, H_{j_2} \in L_{k_1}^0 \cap \mathcal{H}$ such that $(H_{j_1}, H_{i_1}), (H_{i_1}, H_{j_2}) \in L_{k_1}^0$. Now, define L^* such that

$$L_k^* = L_k^0 \quad k \neq k_1, k \neq k_2 \quad k = 1, \dots, n$$

$$L_{k_2}^* = \emptyset$$

$$L_{k_1}^* : V_k^* = V_k^0 \sim H_{i_1}$$

$$E_k^* = E_k^0 \cup (H_{j_1}, H_{j_2}) \sim (H_{j_1}, H_{i_1}) \sim (H_{i_1}, H_{j_2})$$

Then, it directly follows that $[(X^0, Y^0), \mathcal{L}^*, q^0]$ satisfies constraints

(i) - (vi) of problem (2.1) (constraints (ii), (iii), (iv), (vi) are

satisfied as (X^0, Y^0) , q^0 are not changed; the left hand side of (v)

if anything decreases and for (i): $L_{k_2}^* \cap \mathcal{S} = \emptyset \Rightarrow q_{i_{k_2}}^0 = 0 \quad i = 1, \dots, N$.

So $L_{k_2}^* = \emptyset$ does not violate (i), and $|L_{k_1}^0 \cap \mathcal{S}| = 2$, so $L_{k_1}^* \cap \mathcal{S} \neq \emptyset$ and

it does not violate (i)).

Also $z[(X^0, Y^0), \mathcal{L}^*, q^0] \leq z[(X^0, Y^0), \mathcal{L}^0, q^0]$, by triangle inequality, as

$$z[(X^0, Y^0), \mathcal{L}^*, q^0] = z[(X^0, Y^0), \mathcal{L}^0, q^0] + d[(X_{j_1}, Y_{j_1}), (X_{j_2}, Y_{j_2})]$$

$$-d[(X_{j_1}, Y_{j_1}), (X_{i_1}, Y_{i_1})] - d[(X_{i_1}, Y_{i_1}), (X_{j_2}, Y_{j_2})]$$

As $[(X^0, Y^0), \mathcal{L}^*, q^0]$ satisfies (3), the proof is completed. (If

$|L_{k_1}^0 \cap \mathcal{S}| = p > 2$, then the above procedure would be repeated $p - 1$ times).

□

In words Lemma 2.1 says "we can consider only the routings in which a vehicle is either idle ($\mathcal{L}_k^* = \emptyset$) or serves a single supply point ($|\mathcal{L}_k^* \cap \mathcal{S}| = 1$), without any loss of optimality".

LEMMA 2.2: Let $[(X^0, Y^0), \mathcal{L}^0, q^0]$ be a feasible solution to problem (2.1). Then there exists $[(X^0, Y^0), \mathcal{L}^0, q^*]$ feasible for (2.1) and such that

$$\sum_{k=1}^n q_{ik}^* = Q_i \quad i = 1, \dots, N; \quad z[(X^0, Y^0), \mathcal{L}^0, q^*] = z[(X^0, Y^0), \mathcal{L}^0, q^0] \quad (4)$$

PROOF: Define q^* such that:

$$\begin{aligned} &\text{if } H_i \notin L_k \text{ or } L_k \cap \mathcal{S} = \emptyset, \text{ set } q_{ik}^* = q_{ik}^0 = 0 \\ &\text{otherwise} \\ &\text{if } \sum_{h=1}^{k-1} q_{ih}^* + q_{ih}^0 \leq Q_i, \text{ set } q_{ik}^* = q_{ik}^0 \\ &\text{if } \sum_{h=1}^{k-1} q_{ih}^* + q_{ih}^0 > Q_i, \text{ set } q_{ik}^* = Q_i - \sum_{h=1}^{k-1} q_{ih}^* \end{aligned}$$

Then $[(X^0, Y^0), \mathcal{L}^0, q^*]$ is feasible and satisfies (4). □

The meaning of Lemma 2.2 is "we can disregard feasible solution in which some demand points receive more than their requirements, without any loss of optimality".

LEMMA 2.3: Let $[(X^0, Y^0), \mathcal{L}^0, q^0]$ be a feasible solution to problem (2.1), and for which condition (3) and (4) hold. Then there exists

$[(X^0, Y^0), \mathcal{L}^*, q^0]$ such that

$$q_{ik}^0 > 0 \quad \text{if } H_i \in L_k^* \quad i = 1, \dots, N; \quad k = 1, \dots, n \quad (5)$$

and $z[(X^0, Y^0), \mathcal{L}^*, q^0] \leq z[(X^0, Y^0), \mathcal{L}^0, q^0]$ and $[(X^0, Y^0), \mathcal{L}^*, q^0]$ is still feasible with condition (3), (4) holding.

PROOF: Assume that for $L_k^0 \in \mathcal{L}^0$: $q_{ik}^0 = 0$, $H_i \in L_k^0$, $H_i \in \mathcal{D}$. Then there are $H_{v_1}, H_{v_2} \in L_k^0$ such that $(H_{v_1}, H_i), (H_i, H_{v_2}) \in L_k^0$. Define $\mathcal{L}^*$ such that

$$L_j^* = L_j^0 \quad j \neq k \quad j = 1, \dots, n$$

$$L_k^*: V_k^* = V_k^0 \sim \{H_i\}$$

$$E_k^* = E_k^0 \cup (H_{v_1}, H_{v_2}) \sim (H_{v_1}, H_i) \sim (H_i, H_{v_2})$$

Then $[(X^0, Y^0), \mathcal{L}^*, q^0]$ is feasible to problem (2.1) and conditions (3), (4), (5) hold and $z[(X^0, Y^0), \mathcal{L}^*, q^0] \leq z[(X^0, Y^0), \mathcal{L}^0, q^0]$. □

Lemma 2.3 says "we can disregard feasible solutions in which supply vehicles go to some demand points but do not deliver any goods, without any loss of optimality".

COROLLARY 2.3: Let $[(X^0, Y^0), \mathcal{L}^0, q^0]$ be a feasible solution to problem (2.1), and for which condition (3) and (4) holds. Then there exists a $[(X^0, Y^0), \mathcal{L}^*, q^0]$ such that $\mathcal{L}^*$ is a collection of n circuits and $z[(X^0, Y^0), \mathcal{L}^*, q^0] \leq z[(X^0, Y^0), \mathcal{L}^0, q^0]$ and $[(X^0, Y^0), \mathcal{L}^*, q^0]$ is still feasible with conditions (3), (4) holding

PROOF: The proof of this corollary follows very closely the constructive proof of lemma 2.3 and will be omitted □

The above corollary says that, without any loss of optimality, we can disregard feasible solutions in which a vehicle visits a demand point more than once. So from now on we will restrict the Euler graphs

in the feasible solution set to be just circuits. Let us redefine

$\mathcal{L} = \{L_1, \dots, L_n\}$ where $L_i \equiv L_i(V_i, E_i)$ is a circuit, $V_i \in \mathcal{V}$ and it is the route travelled by vehicle i .

LEMMA 2.4: Let $[(X^0, Y^0), \mathcal{L}^0, q^0]$ be a feasible solution to problem (2.1) with condition (3), (4), (5) holding. Then there exists $[(X^0, Y^0), \mathcal{L}^*, q^*]$ such that

$$|L_i^* \cap L_j^* \sim \mathcal{L}| \leq 1 \quad i = 1, \dots, n; j = 1, \dots, n; i \neq j \quad (6)$$

and $z[(X^0, Y^0), \mathcal{L}^*, q^*] \leq z[(X^0, Y^0), \mathcal{L}^0, q^0]$ and $[(X^0, Y^0), \mathcal{L}^*, q^*]$ still feasible with conditions (3), (4), (5) holding

PROOF: Assume $\mathcal{L}^0$ is such that $|L_i^0 \cap L_j^0 \sim \mathcal{L}| = 2$. Then there exists $H_v, H_u \in L_i^0 \cap L_j^0 \cap \mathcal{L}$. As $L_i^0, i = 1, \dots, n$ are Euler graphs, there exists $H_{i_1}, H_{i_2}, H_{i_3}, H_{i_4} \in L_i^0$ and $H_{j_1}, H_{j_2}, H_{j_3}, H_{j_4} \in L_j^0$ such that (see Figure 2.1)

$$(H_{i_1}, H_v), (H_v, H_{i_2}) \in E_i^0$$

$$(H_{i_3}, H_u), (H_u, H_{i_4}) \in E_i^0$$

$$(H_{j_1}, H_v), (H_v, H_{j_2}) \in E_j^0$$

$$(H_{j_3}, H_u), (H_u, H_{j_4}) \in E_j^0$$

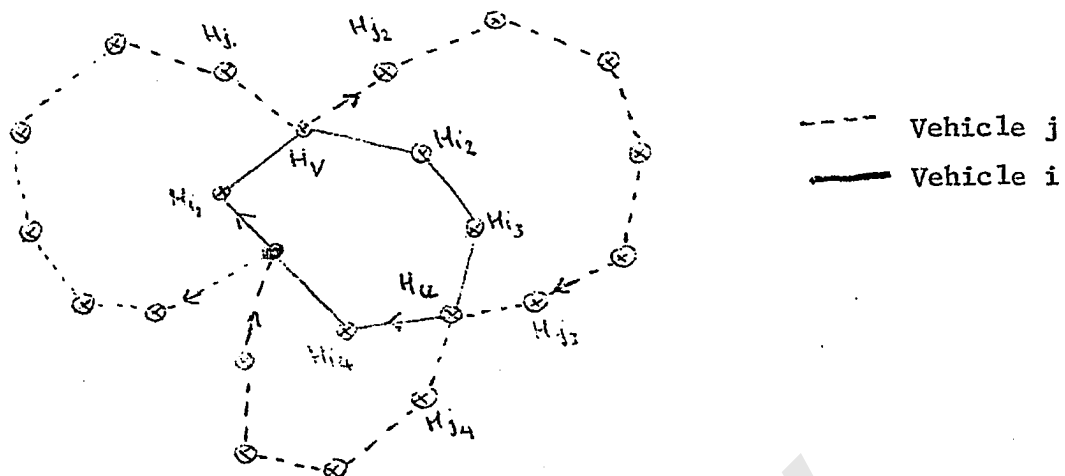


Figure 2.1

Assume $q_{ui}^0 \leq q_{vj}^0$, then define q^* such that

$$\begin{aligned} q_{ui}^* &= q_{ui}^0 - q_{ui}^0 = 0 \\ q_{vi}^* &= q_{vi}^0 + q_{ui}^0 \\ q_{vj}^* &= q_{vj}^0 - q_{ui}^0 \\ q_{uj}^* &= q_{uj}^0 + q_{ui}^0 \\ q_{wz}^* &= q_{wz}^0 \quad \text{otherwise} \end{aligned}$$

Define $\mathcal{L}^*$ such that

$$\begin{aligned} L_i^*: V_i^* &= V_i^0 \sim H_u \\ E_i^* &= E_i^0 \cup (H_{i_3}, H_{i_4}) \sim (H_{i_3}, H_u) \sim (H_u, H_{i_4}) \\ L_k^* &= L_k^0 \quad k = 1, \dots, n \quad k \neq i \end{aligned}$$

Then it directly follows that $[(X^0, Y^0), \mathcal{L}^*, q^*]$ is feasible with conditions (3), (4), (5), (6) holding and $z[(X^0, Y^0), \mathcal{L}^*, q^*] \leq z[(X^0, Y^0), \mathcal{L}^0, q^0]$ (If $|L_i^0 \cap L_j^0 \sim \emptyset| = p > 2$, then the above procedure would be repeated $p - 1$ times).

□

Lemma 2.4 says "we can disregard feasible solutions in which two vehicle routes have two or more demand points in common, without any loss of optimality".

Now define the set $\ell(q)$:

$$\ell(q) = \{L/\forall L_j, L_k \in L: |L_j \cap L_k \cap \mathcal{D}| \leq 1; |L_k \cap \mathcal{D}| = 1; H_i \in L_k \Rightarrow q_{ik} > 0, \\ i = 1, \dots, N\} \quad (7)$$

Also define $\mathcal{C}$ to be the convex hull of (X_i, Y_i) $i = 1, \dots, N$ and $\mathcal{C}_B$ to be the boundary of $\mathcal{C}$. Then,

LEMMA 2.5: Assume $\mathcal{C}_B \in \omega$. Let $[(X^0, Y^0), \mathcal{L}^0, q^0]$ be a feasible solution of problem (2.1) with conditions (3), (4), (5), (6) holding. Then there exists $[(X^*, Y^*), \mathcal{L}^0, q^0]$ such that

$$(X_i^*, Y_i^*) \in \mathcal{C} \cap \omega \quad i = 1, \dots, N + M \quad (8)$$

and $z[(X^*, Y^*), \mathcal{L}^0, q^0] \leq z[(X^0, Y^0), \mathcal{L}^0, q^0]$

PROOF: Define (X^*, Y^*) such that

$$(X_i^*, Y_i^*) = (X_i^0, Y_i^0) \quad \text{if } (X_i^0, Y_i^0) \in \mathcal{C} \\ (X_i^*, Y_i^*) = (\tilde{X}_i, \tilde{Y}_i) \quad \text{if } (X_i^0, Y_i^0) \notin \mathcal{C}$$

where $(\tilde{X}_i, \tilde{Y}_i)$ is such that

$$\delta = d[(\tilde{X}_i, \tilde{Y}_i), (X_i^0, Y_i^0)] \leq d[(w, z), (X_i^0, Y_i^0)] \quad \forall (w, z) \in \mathcal{C} \quad (9)$$

then $(\tilde{X}_i, \tilde{Y}_i) \in \mathcal{C}_B$ and $(X_i^*, Y_i^*) \in \mathcal{C} \cap \omega$, $i = 1, \dots, N + M$. (Recall that (X_i, Y_i) $i = 1, \dots, N$ are constant and are in $\mathcal{C}$ by definition, in

any feasible solution (X, Y) .

Now as condition (3) holds we can partition each E_i^0 in the following two nonintersecting subsets

$$E_i^1 = \{ (H_{i_1}, H_{i_2}) \in E_i^0 / H_{i_1}, H_{i_2} \in \mathcal{C} \}$$

$$E_i^2 = \{ (H_{i_1}, H_{i_2}) \in E_i^0 / H_{i_1} \text{ or } H_{i_2} \notin \mathcal{C} \}$$

Then

$$z[(X^0, Y^0), \mathcal{L}^0, q^0] = \sum_{L_i^0 \in \mathcal{L}^0} \left\{ \sum_{(H_{i_1}, H_{i_2}) \in E_i^1} d[(X_{i_1}^0, Y_{i_1}^0), (X_{i_2}^0, Y_{i_2}^0)] \right. \\ \left. + \sum_{(H_{i_1}, H_{i_2}) \in E_i^2} d[(X_{i_1}^0, Y_{i_1}^0), (X_{i_2}^0, Y_{i_2}^0)] \right\}$$

and

$$z[(X^*, Y^*), \mathcal{L}^0, q^0] = \sum_{L_i^0 \in \mathcal{L}^0} \left\{ \sum_{(H_{i_1}, H_{i_2}) \in E_i^1} d[(X_{i_1}^0, Y_{i_1}^0), (X_{i_2}^0, Y_{i_2}^0)] \right. \\ \left. + \sum_{(H_{i_1}, H_{i_2}) \in E_i^2} d[(X_{i_1}^*, Y_{i_1}^*), (X_{i_2}^0, Y_{i_2}^0)] \right\}$$

So the proof will be completed if we can show that

$$d[(X_\ell, Y_\ell), (\tilde{X}_i, \tilde{Y}_i)] \leq d[(X_\ell, Y_\ell), (X_i^0, Y_i^0)] \quad (10)$$

for $X_\ell \in \mathcal{D}$ and $H_i \in \mathcal{L}$ with $(X_i^0, Y_i^0) \notin \mathcal{C}$.

Consider the angle, θ , made by line segments $c_1 = [(X_i^0, Y_i^0), (\tilde{X}_i, \tilde{Y}_i)]$ and $c_2 = [(\tilde{X}_i, \tilde{Y}_i), (X_\ell, Y_\ell)]$. If $\theta < 90^\circ$, then as $c_2 \in \mathcal{C}$ condition (9) would be violated, therefore $\theta \geq 90^\circ$. But then (10) holds by rules of simple geometry.

□